

Applied Optimization

Exercise 2 - Convex Functions

September 18, 2025

What to hand in

A .zip compressed file renamed to `Exercise $n$ -Group.zip` where n is the number of the current exercise sheet. It should contain:

- Hand in **only** the files you changed (headers and source). It is up to you to make sure that all files that you have changed are in the zip.
- A `readme.txt` file containing a description on how you solved each exercise (use the same numbers and titles) and the encountered problems. Indicate what fraction of the total workload each project member contributed.
Failure to do so will result in you losing a point.
- Other files that are required by your `readme.txt` file. For example, if you mention some screenshot images in `readme.txt`, these images need to be submitted too.
- For the theory exercise, put `TheoryExercise.pdf` with your solutions in the same .zip you submit for the code.
- Submit your solutions to ILIAS before the submission deadline. Late submissions will receive 0 points!
- **Important:** Your submission MUST compile without any issue in order to obtain a passing grade.

Convex Functions (1 pt)

Simple functions (0.5 pt)

Are the following functions convex? Prove your statement.

1. $f(x) = x^2, x \in \mathbb{R}$
2. $f(x) = e^{x^2}, x \in \mathbb{R}$
3. $f(x, y) = x^2 + 3xy + 2y^2, x \in \mathbb{R}, y \in \mathbb{R}$

Log-sum-exp (0.5 pt)

Show that the function

$$f(\mathbf{x}) = \log(e^{x_1} + \dots + e^{x_n})$$

is convex on $\mathbb{R}^n$.

Hint: The proof is outlined in the Boyd's book p. 74. Develop by it yourself and provide the intermediate steps.

Geometric mean (Bonus (0.5 pt))

Show that the geometric mean

$$f(\mathbf{x}) = (\prod_{i=1}^n x_i)^{1/n}$$

is concave on $\mathbb{R}_{++}^n$.

Programming Exercise: Convexity Test (1 pt)

By definition, a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if $\text{dom} f$ is a convex set and if for all $x, y \in \text{dom} f$, and θ with $0 \leq \theta \leq 1$, we have

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y).$$

One can show the convexity of a function by theoretical proof or by simply running a brute-force check of the above condition until you find a counter-example. Of course, if the function really is convex, you won't be able to find such a counter-example. The best you can do is to "guess" that the function is convex, based on a high number of trials. In this task, you are requested to implement such a search algorithm to check if the functions from the introductory exercise are convex.

The idea of the search algorithm is to randomly sample pairs of points on the domain and check if the convex condition as described is satisfied. To do so, you will have to check

that regularly sampled points between those two random points satisfy the convexity condition. i.e. the two points x and y should be random, but the θ giving the convex combination of those two points should be increased by a fixed delta $n_evals-1$ times.

You need to implement the function `isConvex()` in `Algorithms/ConvexityTest.hh`. You can use the function `RandomNumberGenerator::get_random_nd_vector(n)` to generate random sampling points.

The algorithm should either stop when a counter-example point is found, or after a fixed number of sampling points have been tried, in which case the function is considered convex.

Again, you can use the `ConvexityTest-test` executable to test your implementation. As usual, passing the tests is not a sufficient condition to obtain a full grade.